RUTUJA KADAM

DAY 05 – ETHERNET LAN SWITCHING FUNDAMENTALS

THEORY (10 Marks)

1. What is Ethernet?

Ethernet is a widely used networking technology for wired Local Area Networks (LANs). It defines how devices communicate with each other over Ethernet networks. It is commonly used in homes, offices, schools and data centers.

2. What is an Ethernet frame?

An Ethernet frame is the unit of data transmitted across an Ethernet network. It contains the following fields:

1. Destination MAC Address – Identifies the intended receiver.
2. Source MAC Address – Identifies the sender.
3. Type/Length – Indicates the protocol carried in the payload.
4. Payload – Contains the actual data.
5. Frame Check Sequence (FCS) – Helps detect transmission errors.

3. What is a MAC address?

A MAC (Media Access Control) address is a unique identifier assigned to a network interface. It is used for communication at Layer 2 of the OSI Model.

Example: 00:1A:2B:3C:4D:5E.

4. What does MAC stand for?

MAC stands for **Media Access Control**. It is used to identify a network interface at the Data Link Layer (Layer 2).

5. Difference between a MAC address and an IP address?

MAC Address	IP Address
MAC stands for Media Access Control.	IP stands for Internet Protocol.
It works at Layer 2.	It works at Layer 3.
It identifies a network interface.	It provides logical addressing to a device.
It is used for local network communication.	It is used for communication across different networks.
Example: 00:1A:2B:3C:4D:5E	Example: 192.168.1.10

6. What is a MAC Address Table?

A MAC Address Table is a table maintained by a network switch that stores MAC addresses and their corresponding switch ports. It helps the switch identify which port should receive a particular Ethernet frame.

7. How does a switch learn MAC addresses?

A switch learns MAC addresses through a process called Dynamic MAC Learning.

The process is:

1. A device sends an Ethernet frame to the switch.
2. The switch receives the frame on a particular port.
3. It examines the source MAC address.
4. It learns which device is connected to that port.
5. It stores the MAC address and port information in its MAC Address Table.

8. What is a source MAC address?

A source MAC address identifies the network interface that sent the Ethernet frame. A switch examines the source MAC address to learn which device is connected to which port.

9. What is a destination MAC address?

A destination MAC address identifies the intended receiver of an Ethernet frame. A switch examines the destination MAC address to decide where to forward the frame.

10. What is an unknown unicast frame?

An unknown unicast frame is an Ethernet frame whose destination MAC address is not present in the switch's MAC Address Table. Since the switch does not know the correct destination port, it floods the frame through other ports in the same Layer 2 domain.

11. What is flooding?

Flooding is the process in which a switch forwards an Ethernet frame through all other ports in the same Layer 2 domain, except the port on which the frame was received. It happens when the destination MAC address is unknown.

12. What is known unicast forwarding?

Known unicast forwarding is the process in which a switch forwards an Ethernet frame only to the specific port associated with the destination MAC address in its MAC Address Table.

13. At which OSI layer does Ethernet switching primarily operate?

Ethernet switching primarily operates at the **Data Link Layer (Layer 2)** of the OSI Model. It uses MAC addresses to forward Ethernet frames between devices.

14. Why does a switch maintain a MAC Address Table?

A switch maintains a MAC Address Table to:

1. Store MAC addresses and their corresponding ports.
2. Identify connected devices.
3. Forward frames to the correct destination port.
4. Avoid unnecessary flooding when the destination is known.
5. Improve network communication efficiency.

15. Why is Ethernet knowledge important in cybersecurity?

Ethernet knowledge is important in cybersecurity because:

1. It helps analyse MAC addresses and Ethernet frames.
2. It helps understand local network traffic.
3. It helps investigate switch behaviour and forwarding decisions.
4. It helps analyse ARP tables and packet captures.
5. It helps understand network segmentation.

6. It supports the identification of suspicious local network traffic and Layer 2 attacks.
7. It provides a foundation for tools like Wireshark and Nmap.

This knowledge is useful in network monitoring, incident response, vulnerability assessment and penetration testing.

PRACTICAL IDENTIFICATION (4 Marks)

Network Details:

1. **Network Adapter Name:** Wireless LAN adapter Wi-Fi (MediaTek Wi-Fi 6E MT7922)
2. **MAC Address (Physical Address):** 14-AC-60-97-E6-55
3. **IPv4 Address:** 192.168.0.101
4. **Default Gateway:** 192.168.0.1

SCREENSHOTS:-

1)ipconfig/all



```
C:\Users\HP>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-GQ12MPJH
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 16-AC-60-97-C6-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 16-AC-60-97-D6-65
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : MediaTek Wi-Fi 6E MT7922 (RZ616) 160MHz PCIe Adapter
   Physical Address. . . . . . . . . : 14-AC-60-97-E6-55
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::49b1:753c:2f33:8f5e%8(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.0.101(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 11 November 2026 20:15:35
   Lease Expires . . . . . . . . . . : 12 November 2026 02:46:36
   Default Gateway . . . . . . . . . : 192.168.0.1
   DHCP Server . . . . . . . . . . . : 192.168.0.1
   DHCPv6 IAID . . . . . . . . . . . : 85240928
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-46-78-B6-48-EA-62-93-48-CB
   DNS Servers . . . . . . . . . . . : 192.168.0.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 14-AC-60-97-E6-56
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek Gaming GbE Family Controller
   Physical Address. . . . . . . . . : 48-EA-62-93-48-CB
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Users\HP>
```

2)getmac

```
C:\Users\HP>getmac

Physical Address    Transport Name
=================== ============================================================
48-EA-62-93-48-CB   Media disconnected
14-AC-60-97-E6-55   \Device\Tcpip_{6085FB92-EDE3-4627-BEA5-B20769146A93}
14-AC-60-97-E6-56   Media disconnected
```

3)arp-a

```
C:\Users\HP>arp -a

Interface: 192.168.0.101 --- 0x8
  Internet Address      Physical Address      Type
  192.168.0.1           f4-f2-6d-2a-67-ec     dynamic
  192.168.0.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\HP>
```